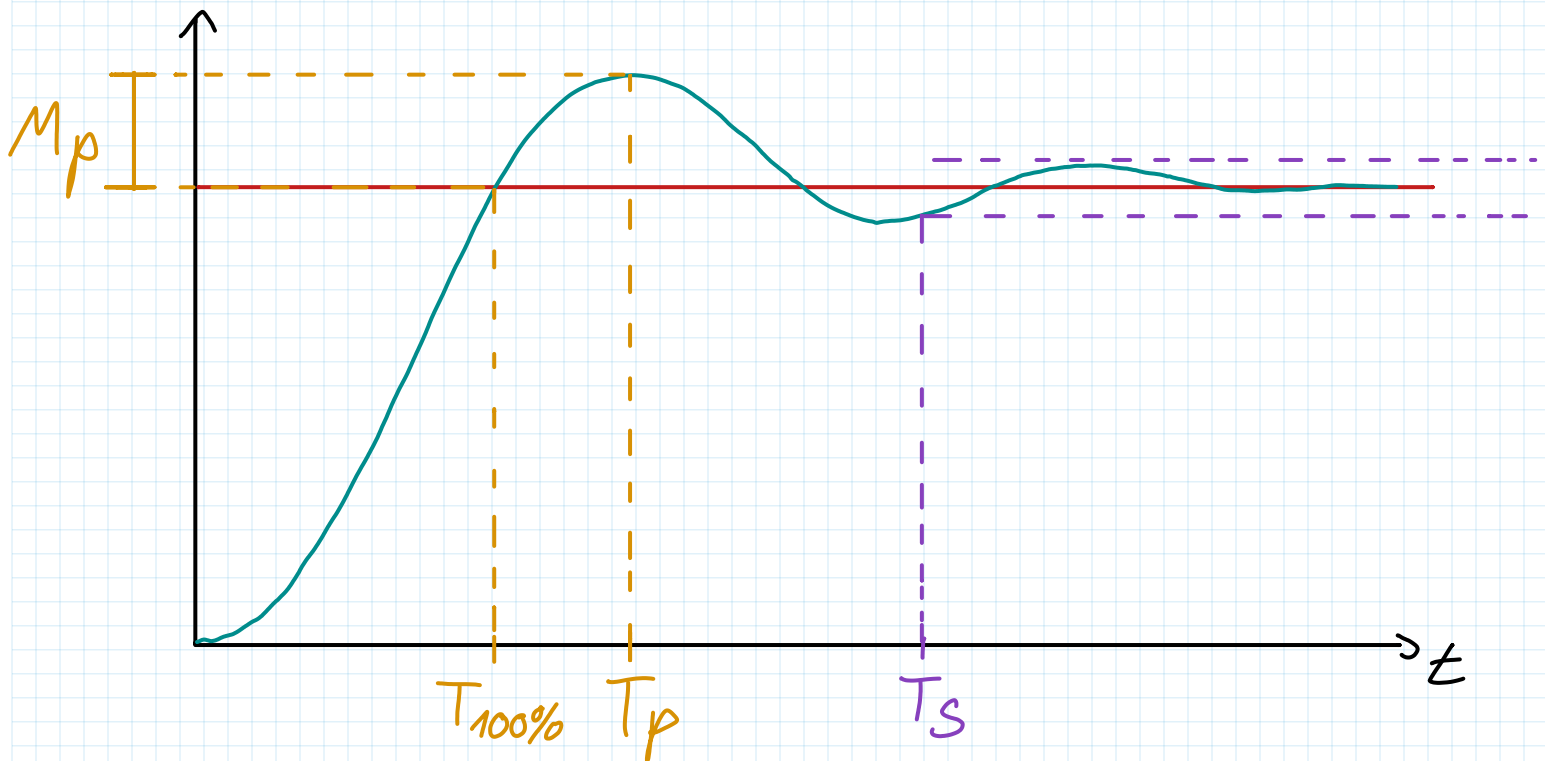


• Last week we looked at time-domain specifications:



• This means we analyzed the response to a test signal in time domain.

⇒ Signal behaviour over TIME.

• But physical systems are in their nature often oscillatory, and excitations also tend to take sinusoidal shapes. ⇒ Step-response analysis will not suffice and we need to analyze...

⇒ Signal behaviour over FREQUENCY

Engineering student classic: Tacoma-Narrows bridge

For this we introduce a new plot...

The Bode Plot

Quick frequency domain recap:

- Many signals can be written as a linear combination of complex exponentials:

$$u(t) = \sum_i U_i e^{s_i t}$$

Example: $u(t) = \cos(\omega t) = \frac{1}{2} e^{j\omega t} + \frac{1}{2} e^{-j\omega t}$

Applying $u(t)$ to an LTI-system $G(s)$, we get:

$$y_{ss}(t) = \sum_i U_i G(s_i) e^{s_i t}, \quad \text{so here we get}$$

$$y_{ss}(t) = \frac{1}{2} G(j\omega) e^{j\omega t} + \frac{1}{2} G(-j\omega) e^{-j\omega t}$$

$\Rightarrow$ hard to interpret because of complex part! Idea:

Portray complex value in polar form with

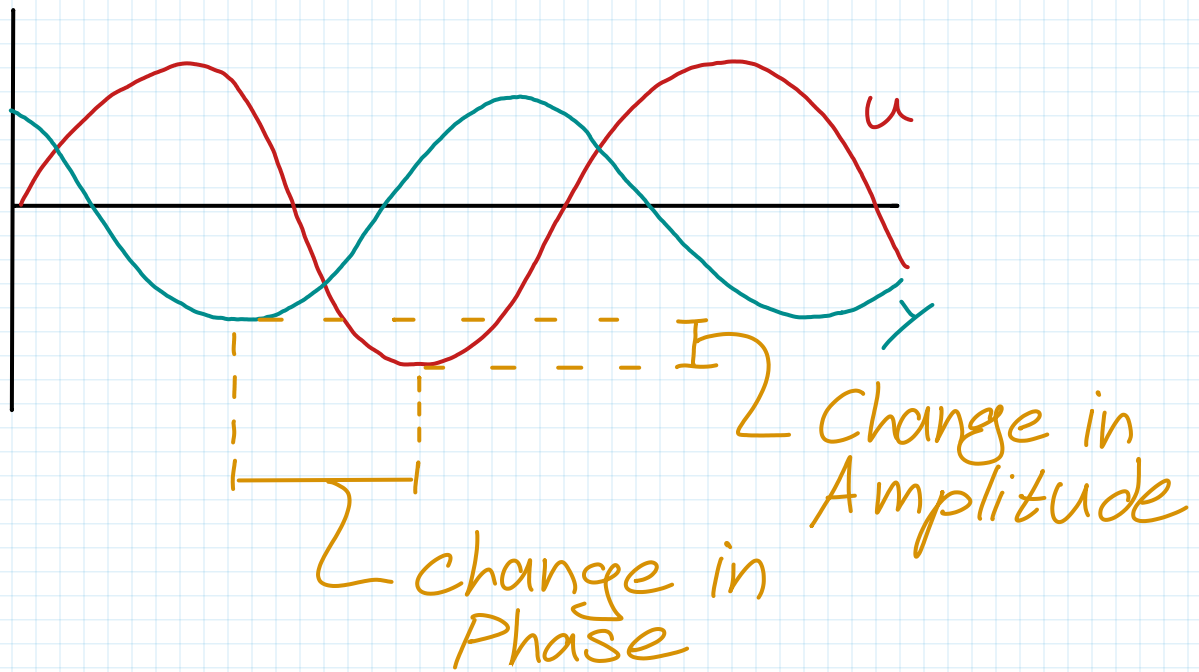
$M(\omega) = |G(j\omega)|$, $\varphi(\omega) = \angle G(j\omega)$, with which we can write a complex number as $M e^{j\varphi}$. Applying this...

$$y_{ss} = \frac{1}{2} \overbrace{M e^{j\varphi}}^{G(j\omega)} e^{j\omega t} + \frac{1}{2} \overbrace{M e^{-j\varphi}}^{G(-j\omega)} e^{-j\omega t}$$

$$= \frac{1}{2} M (e^{j(\omega t + \varphi)} + e^{-j(\omega t + \varphi)})$$

$$= M \cos(\omega t + \varphi) \quad (\text{input was } u(t) = \cos(\omega t))$$

$\Rightarrow M$ and φ depend on ω and fully characterize response!



• Important: For a given input with frequency ω , the steady-state response of an LTI-system has a constant

$$\cdot M(\omega) = |G(j\omega)|,$$

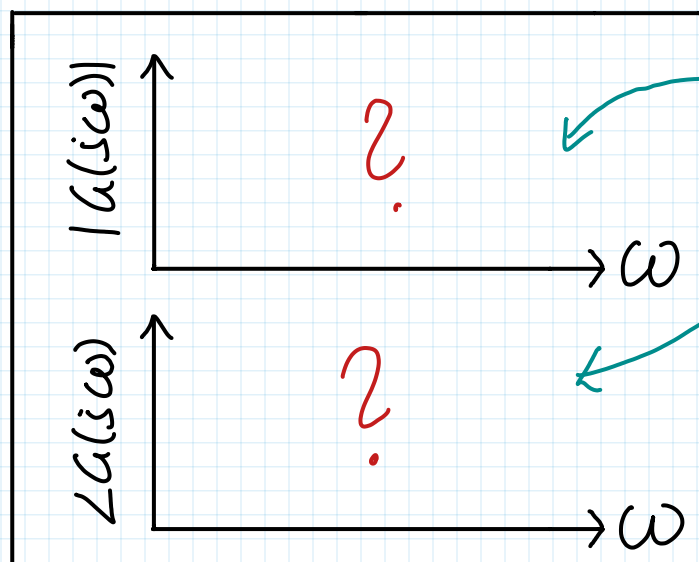
$$\cdot \varphi(\omega) = \angle G(j\omega),$$

and the frequency of the output is always also ω .

This is exactly what the Bode plot shows: Amplitude and phase as a function of ω .

- The Bode plot therefore consists of two graphs as such:

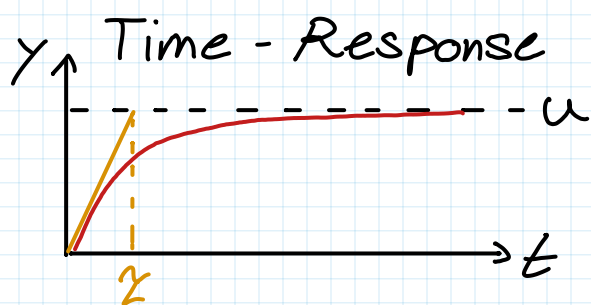
Bode Plot



- "Bode Plot": Both plots
- "Magnitude Plot"
- "Phase Plot"

What do you think Bode plots look like?

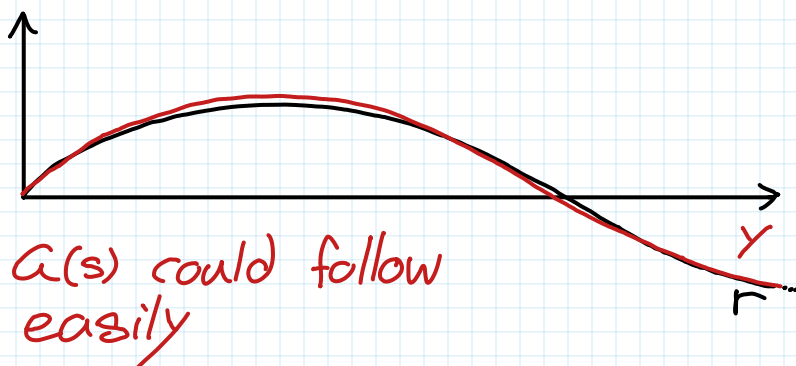
Building intuition:



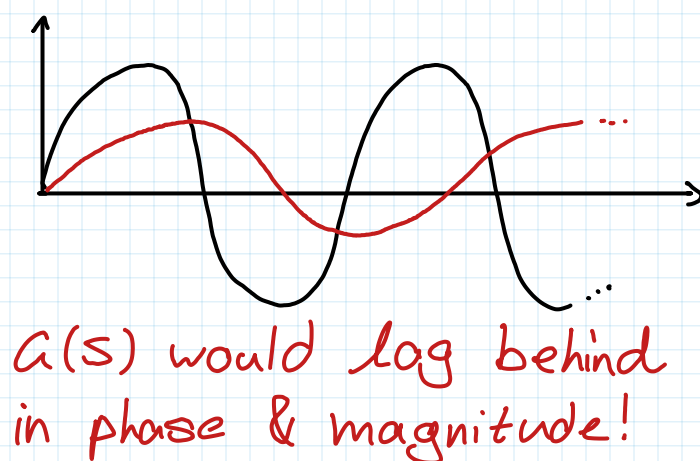
Time constant τ characterizes response of 1st. order system $G(s) =$

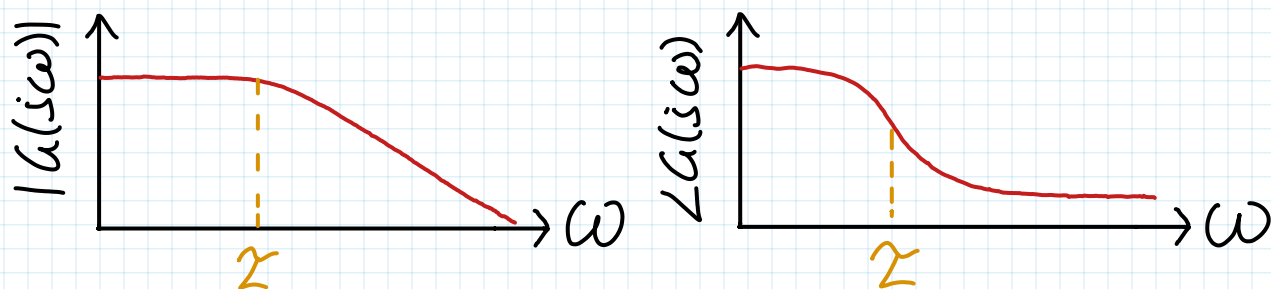
$\Rightarrow$ How fast is $G(s)$?

- Following low ω



- Following high ω





Let's quickly discuss, why bode plots look the way they do:

1) Scaling of vertical axis

- Out of convention, the y-axis of the magnitude plot is scaled in decibels [dB].

$$|G(j\omega)|[\text{dB}] = 20 \cdot \log_{10}(|G(j\omega)|)$$

- The phase plot is shown in degrees [°].

2) Scaling of horizontal axis

- The plots have a logarithmically scaled x-axis in ω .

⇒ The log-log nature of the magnitude plot makes it hard to get intuition..

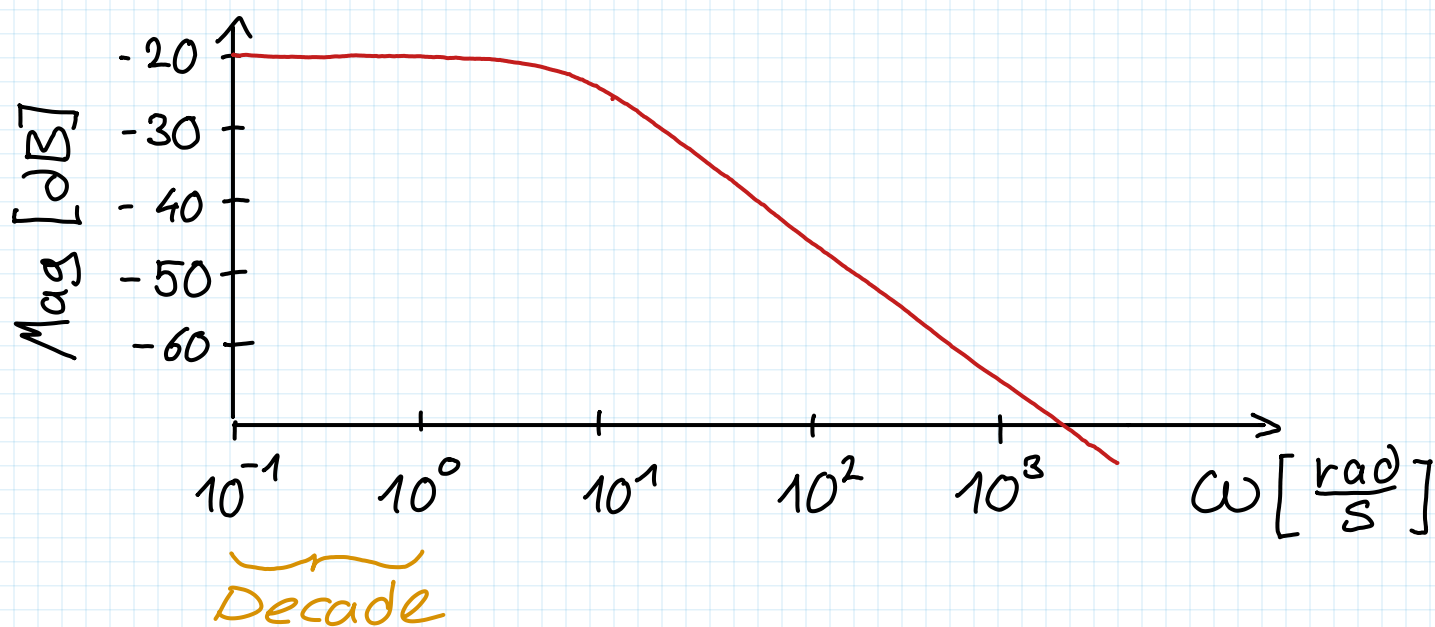
You are given the following system:

$$G(j\omega) = \frac{1}{s + 10} = \frac{1}{j\omega + 10}$$

• First we evaluate the magnitude plot:

$$|G(j\omega)| \text{ [dB]} = 20 \cdot \log_{10} \left(\frac{|1|}{|j\omega + 10|} \right)$$

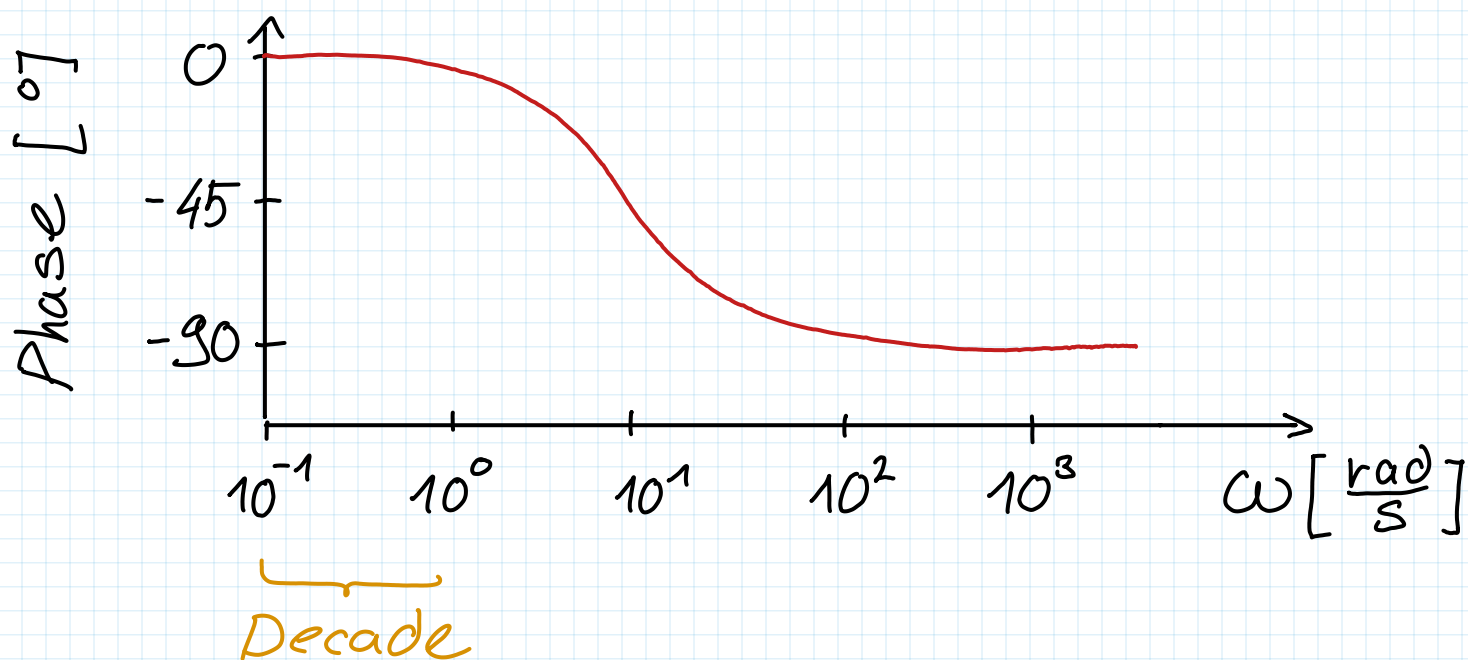
$$= 20 \log_{10} \left(\frac{1}{\sqrt{\omega^2 + 10^2}} \right). \quad \text{Plotting this:}$$



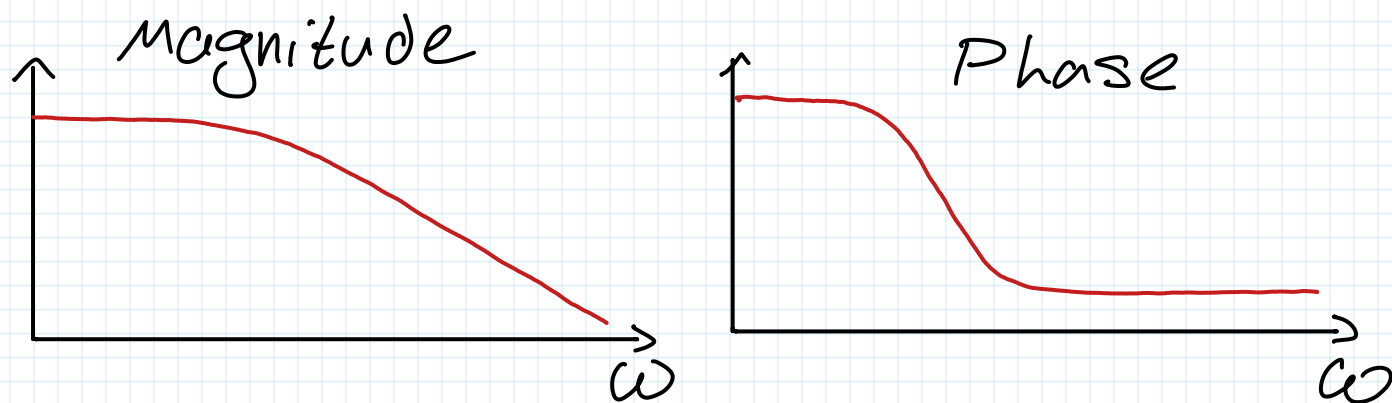
Before we discuss how we got to this, let's look at the phase plot:

• Evaluating the phase:

$$\angle G(j\omega) = \angle \frac{1}{j\omega + 10} = \cancel{\angle 1} - \angle(j\omega + 10) \\ = -\arctan\left(\frac{\omega}{10}\right). \quad \text{Plotting this:}$$



So how can we interpret these plots?
What do they tell us about a system?



This is a very typical Bode configuration. Systems with this behaviour are also called low-pass-filters

So how do we sketch them?

⇒ As usual, we analyze the poles and zeros:

Phase \ Magnitude	Magnitude	
	-20 dB/dec	+20 dB/dec
-90°	stable pole	non-minimum phase zero
+90°	unstable pole	minimum phase zero

They tell us how phase/gain changes over different frequencies.

Let's just try an example!

You are given a system $G(s)$

$$G(s) = \frac{(s+1)(s-100)}{(s+10)(s+30)(s-1000)}$$

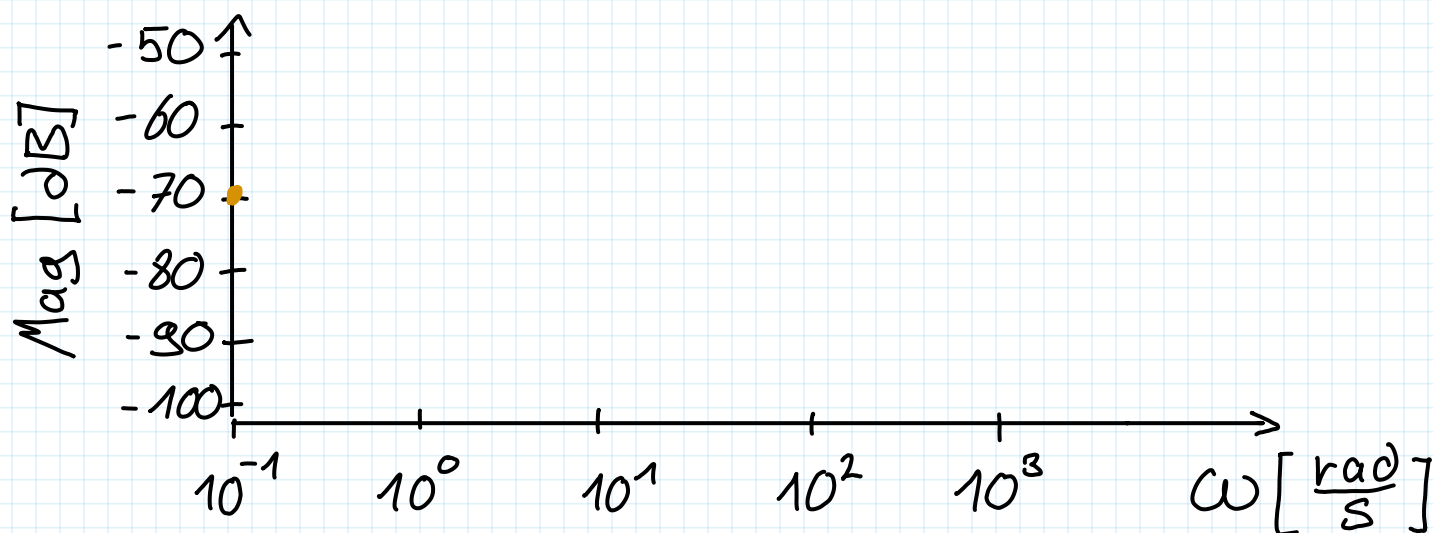
you would like to know how the steady-state frequency response looks like. $\Rightarrow$ Bode!

1) Find Bode-gain $G(0)$

$$G(0) = \frac{1 \cdot (-100)}{10 \cdot 30 \cdot (-1000)} = 3,3 \cdot 10^{-4}$$

$\Rightarrow$ Transform to [dB]

$$G(0)[\text{dB}] = 20 \cdot \log_{10}(3,3 \cdot 10^{-4}) \approx -70$$



2) Sort poles & zeros from smallest to largest (absolute value):

$$G(s) = \frac{(s+1)(s-100)}{(s+10)(s+30)(s-1000)}$$

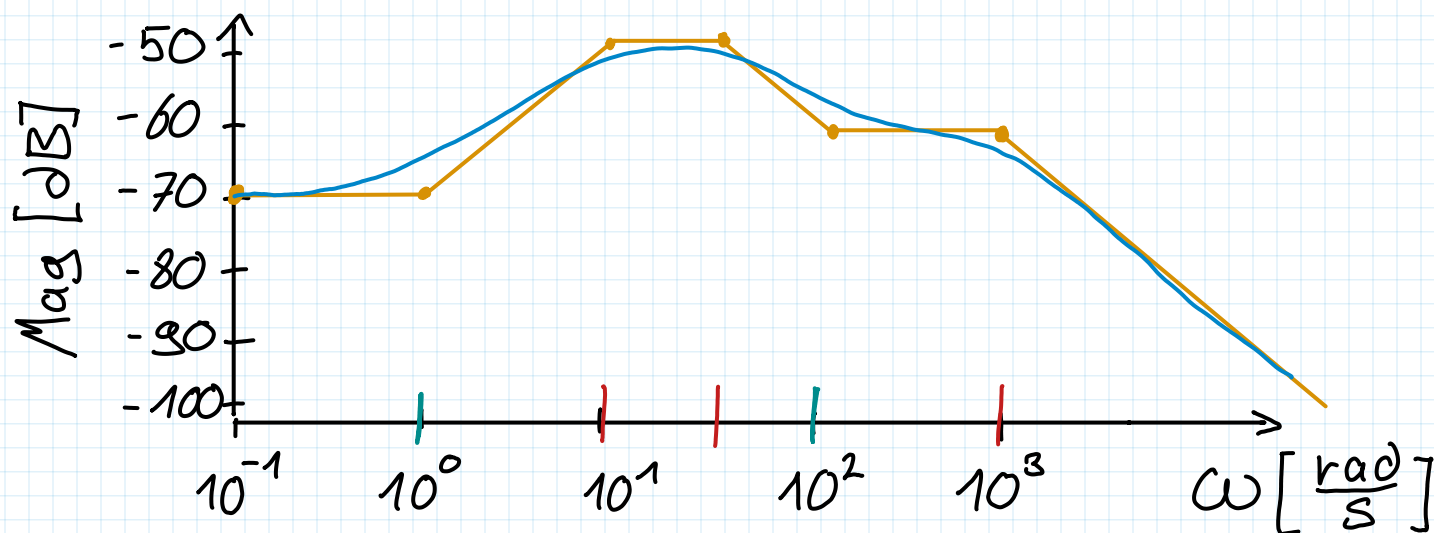
smallest to largest (|·|) →

$$|z_1 = -1, p_1 = -10, p_2 = -30, z_2 = 100, p_3 = 1000|$$

$$z_1 = 1, p_1 = 10, p_2 = 30, z_2 = 100, p_3 = 1000$$

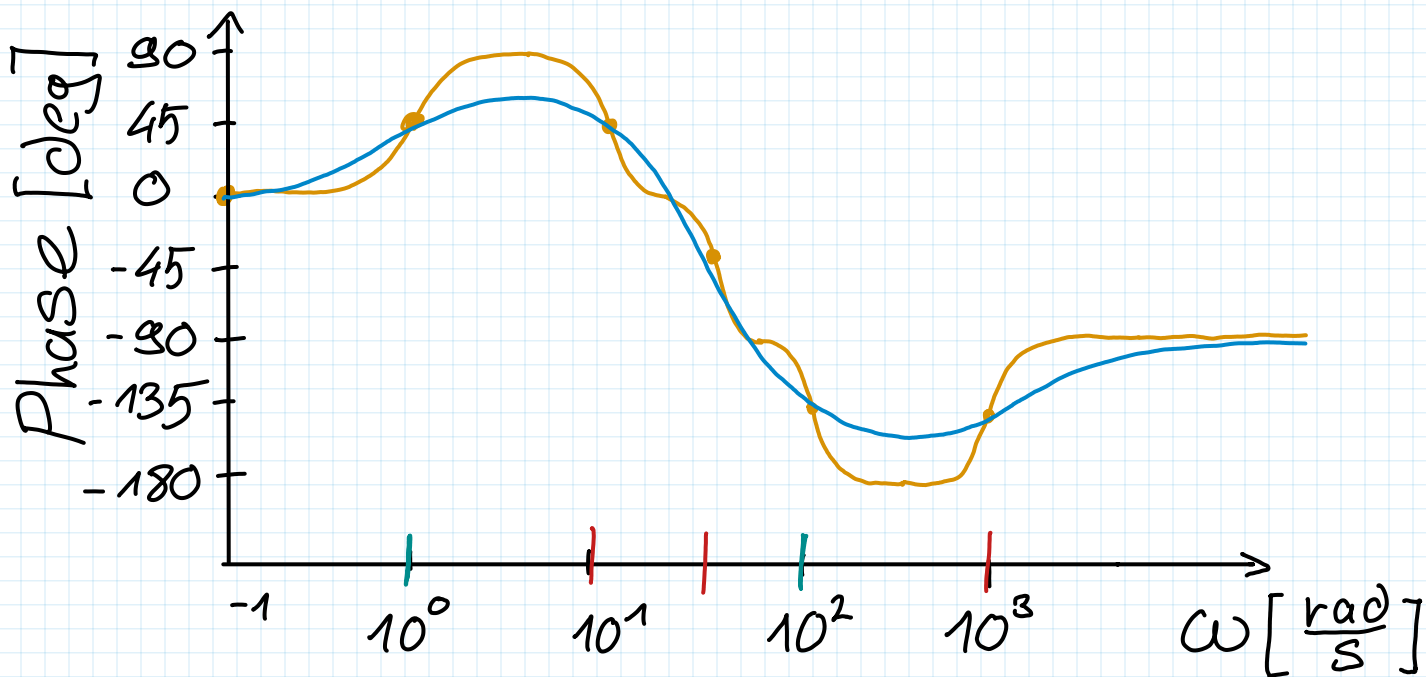
3) We follow the rules and change our steepness whenever we "encounter" a **pole** or **zero**

Magnitude		-20 dB/dec	+20 dB/dec
Phase	-90°	stable pole	non-minimum phase zero
	+90°	unstable pole	minimum phase zero



Now for the phase plot:

We change the phase around a pole/zero.
The change is strongest at the pole/zero.



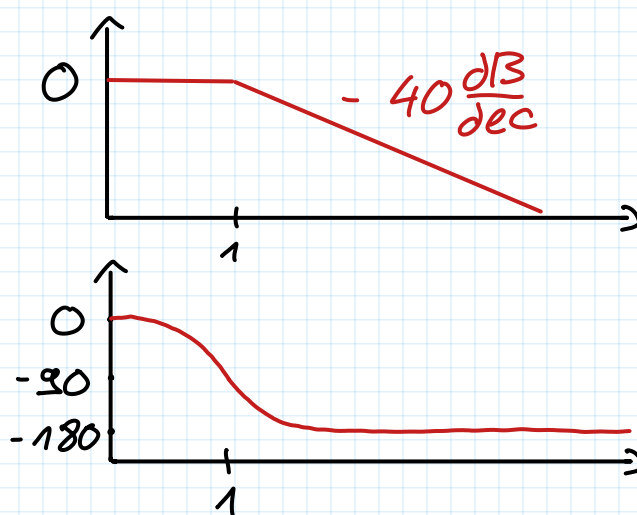
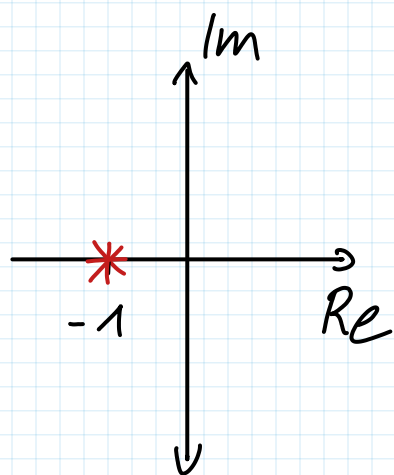
Special rules:

- If tf has pole $p=0$ (integrator), the plot will start with -20dB/dec and -90°
- If tf has zero $z=0$ (differentiator), the plot will start with $+20\text{dB/dec}$ and $+90^\circ$

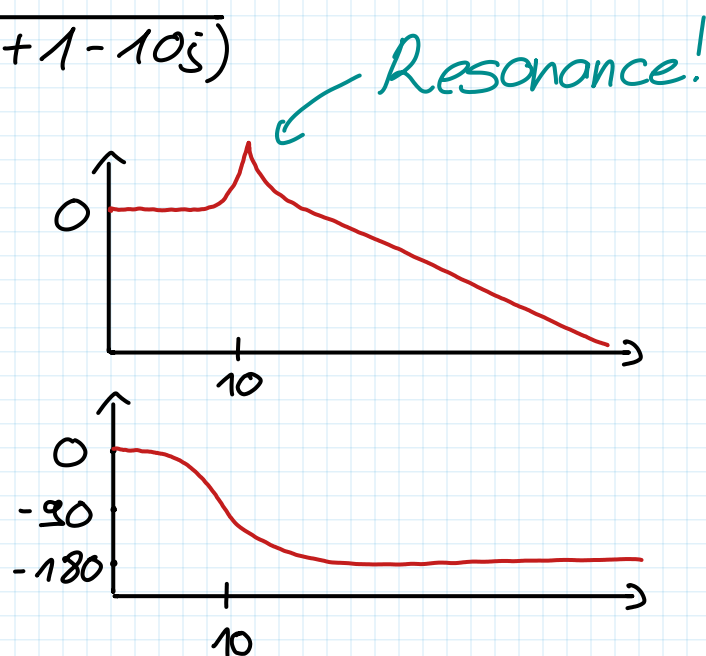
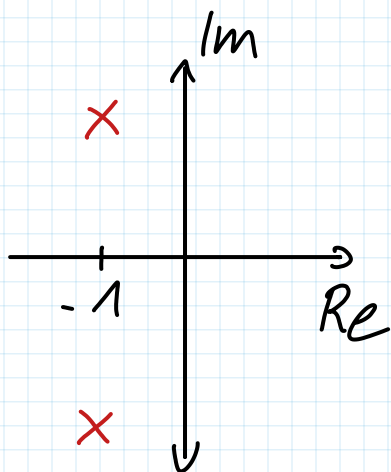
What if we have complex poles/zeros?

Let's compare the following systems:

$$G_1(s) = \frac{1}{(s+1)(s+1)}$$



$$G_2(s) = \frac{1}{(s+1+10j)(s+1-10j)}$$

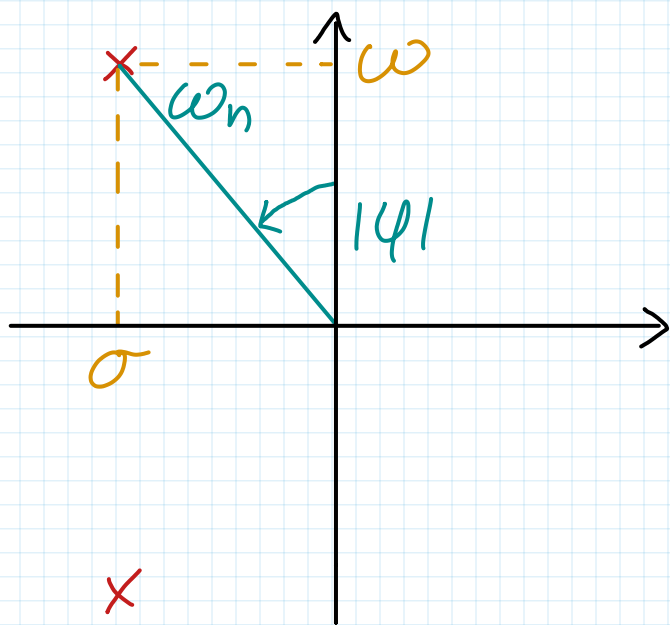


How can we characterize this peak?

• Remember how we wrote second order systems:

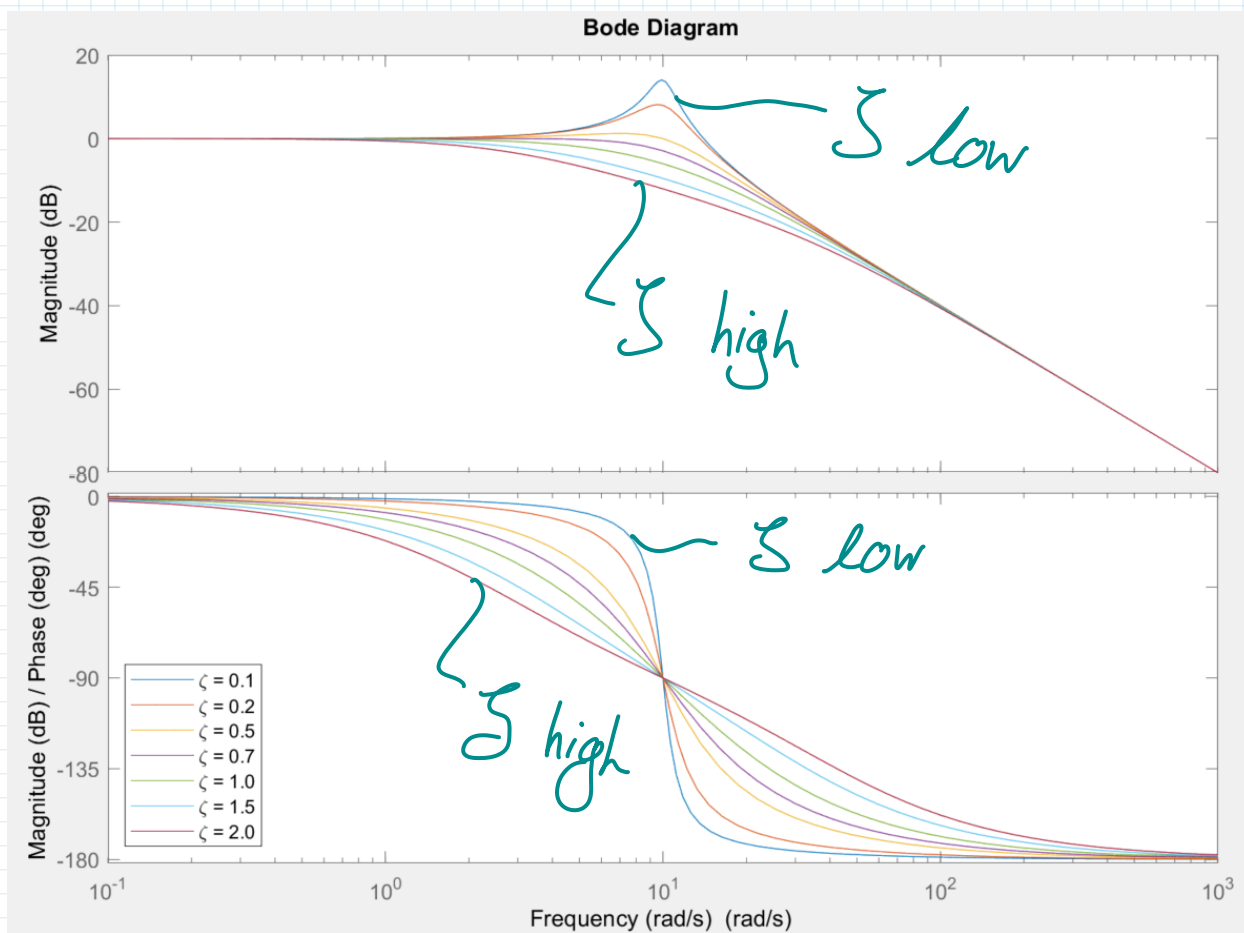
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

It turns out, that the damping factor ζ defines the intensity of the resonance!



- $\omega_n = \sqrt{\sigma^2 + \omega^2}$
- $\varphi = \tan^{-1}\left(\frac{\sigma}{\omega}\right)$
- $|\sigma| = \sin(|\varphi|) \cdot \omega_n$
- $\sin(|\varphi|) = \zeta$

The stronger the damping, the smaller the resonance!

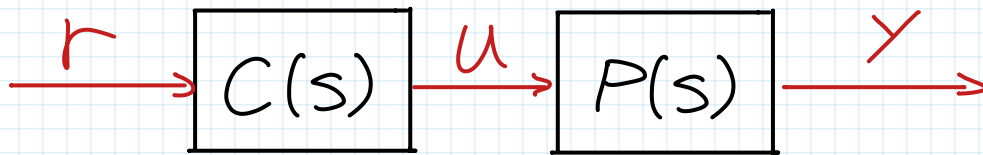


$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

• Note: this is for 2nd order systems, but the general principles stays the same for higher order systems.

• The Bode plot has one more great feature...

Given two systems, controller and plant:



• Magnitudes multiply!

$$|L(s)| = |C(s) \cdot P(s)| = |C(s)| \cdot |P(s)|$$

• Phases add!

$$\angle L(s) = \angle C(s) \cdot P(s) = \angle C(s) + \angle P(s)$$

Remember logarithmic rules:

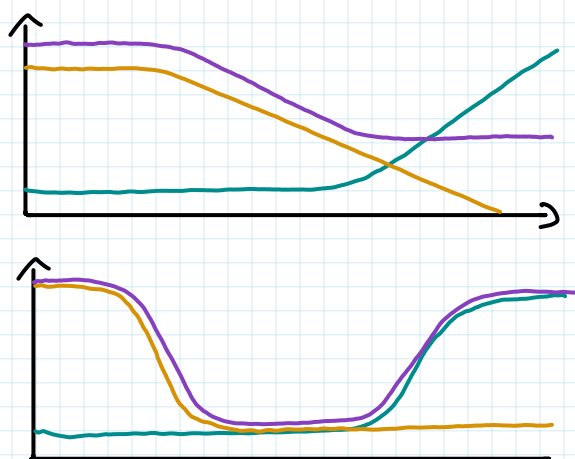
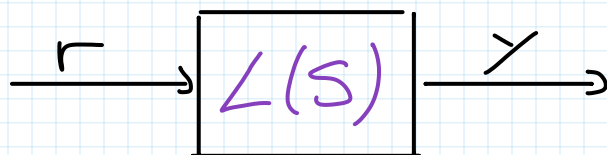
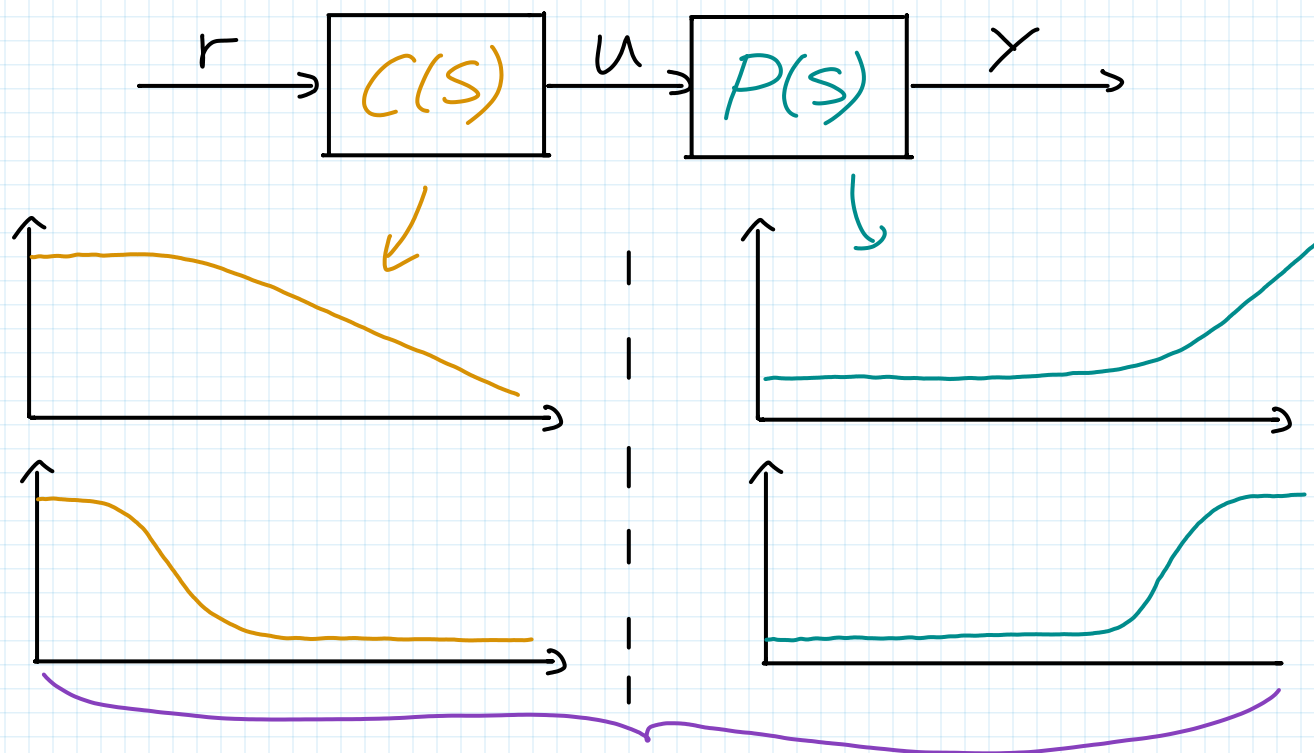
$$\log(A \cdot B) = \log(A) + \log(B)$$

• The magnitude plot is logarithmic:

$$\begin{aligned} \log(|L(s)|) &= \log(|C(s)| \cdot |P(s)|) \\ &= \log(|C(s)|) + \log(|P(s)|) \end{aligned}$$

Thanks to this, we can just superpose Bode plots!

• If you have the Bode plot of $C(s)$ and $P(s)$, you can find the plot of $L(s)$ by adding them on top of each other!



- Thanks to this additive feature, we can also break down complicated tf's into simpler ones and recombine only the plots.

$$B(G(s)) \hat{=} \text{Bode plot of } G(s)$$

$$\cdot G(s)$$

$\Rightarrow$ Draw 1 plot

$$B(G)$$

$$\cdot G(s) = G_1(s) \cdot G_2(s) \cdot G_3(s)$$

$\Rightarrow$ Draw 3 plots

$$B(G_1), B(G_2), B(G_3)$$

$\Rightarrow$ Add them together

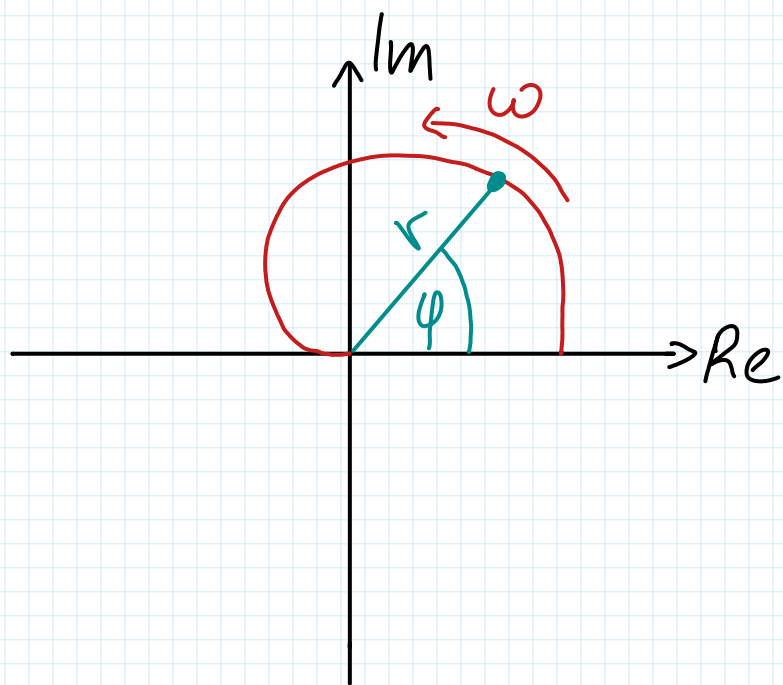
$$B(G) = B(G_1) + B(G_2) + B(G_3)$$

$\Rightarrow$ Same result, you can do it however you prefer.

What the Bode-plot represents can actually also be represented in just one plot...

The polar plot

- In this plot we combine the magnitude and phase into a complex number.
- The frequency is no longer an axis, but implicit, just like the gain k in the root locus

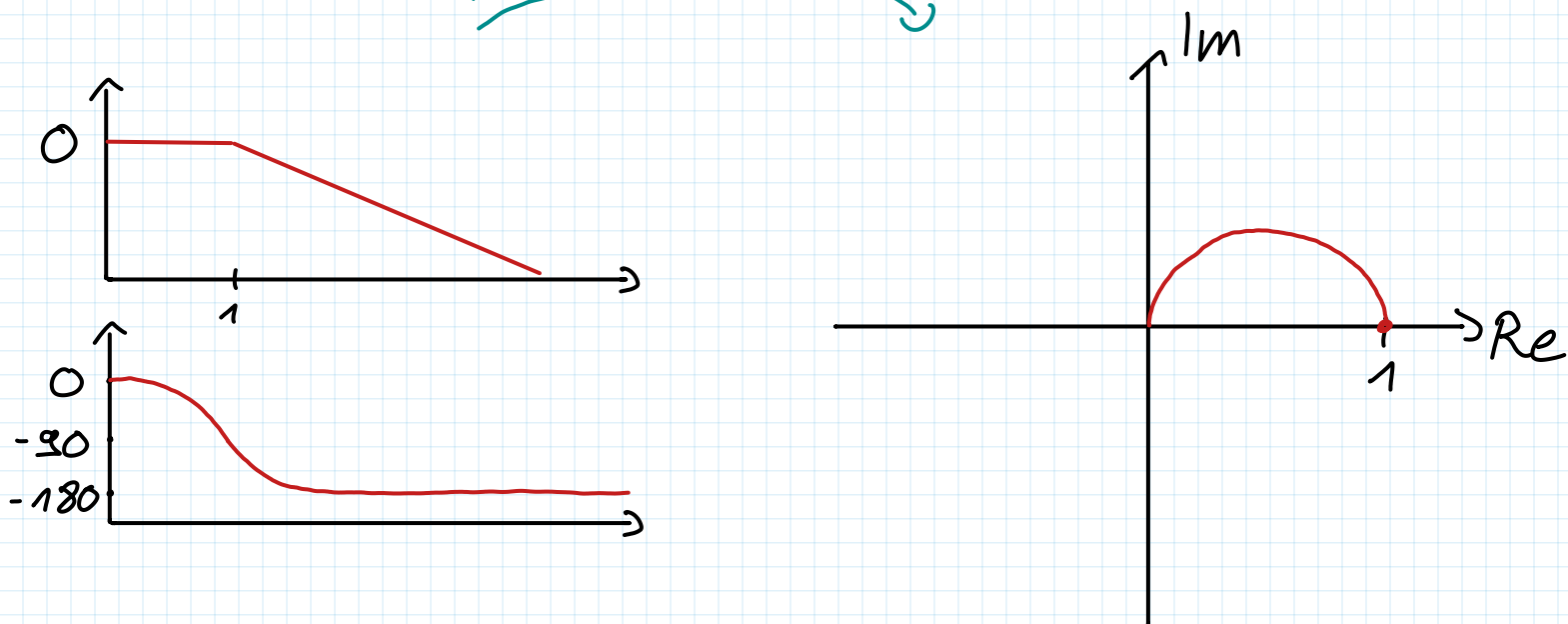


$$\varphi = \angle G(j\omega)$$

$$r = |G(j\omega)|$$

If we have the Bode plots, we extract the polar plot as such:

Bode to Polar



careful: $0\text{dB} = 1$

Note: For all of these plots we are sketching. you won't have to precisely draw them.

Ex: Find the Bode & Polar plots of $G(s)$

$$G(s) = \frac{(s-1)}{(s+10)(s+100)}$$

• First, identify poles/zeros

$$z_1 = 1, \quad p_1 = -10, \quad p_2 = -100$$

• Evaluate Bode gain:

$$G(0) = \frac{-1}{10 \cdot 100} = -1 \cdot 10^{-3}$$

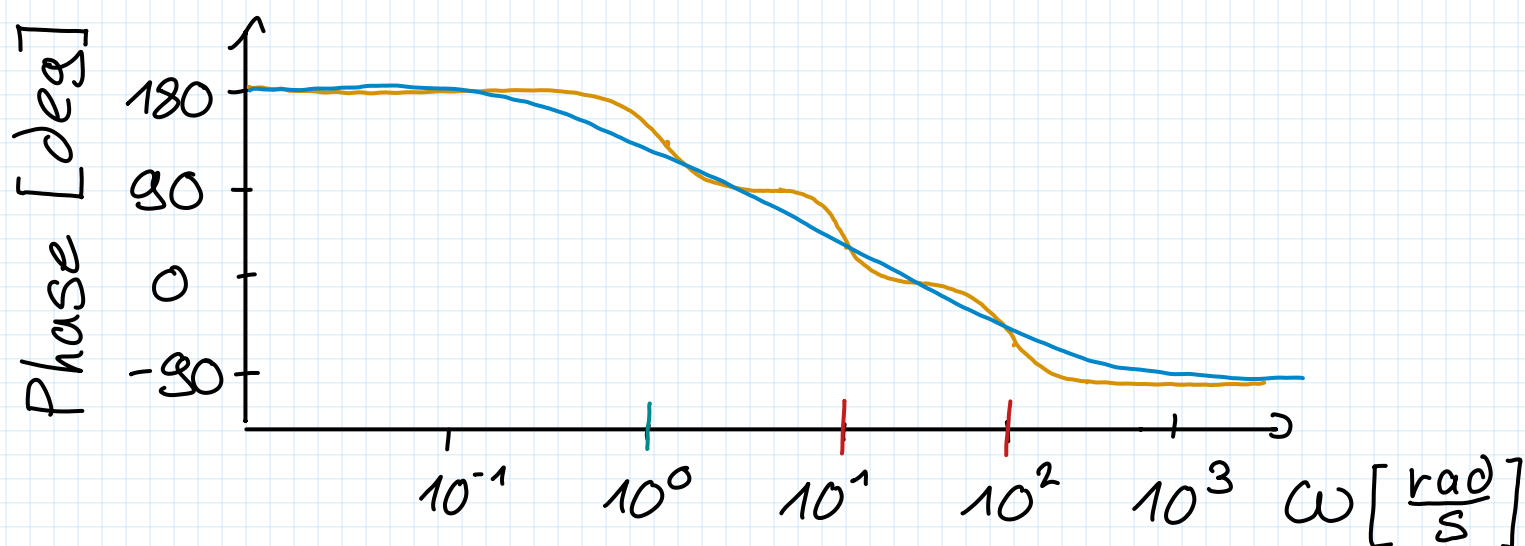
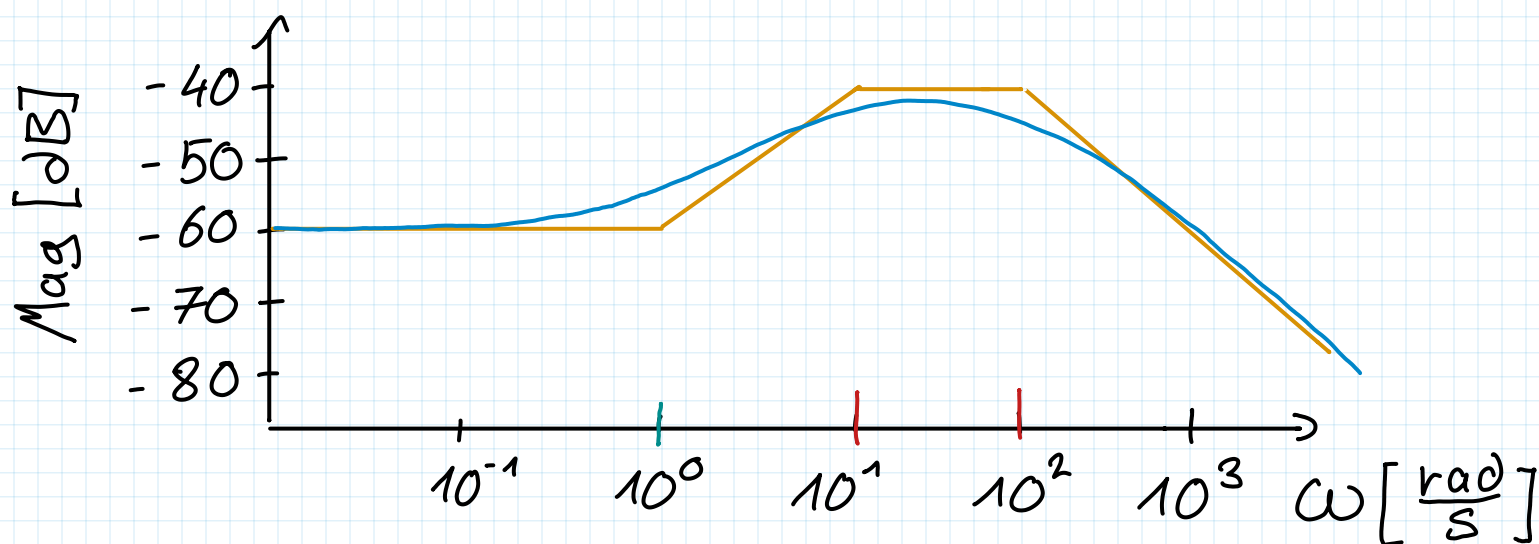
$$\angle(G(0)) = \pi \quad (\text{Phase plot starts at } 180^\circ)$$

$$|G(0)|[\text{dB}] = 20 \cdot \log_{10}(|G(0)|) \approx -60 \text{ dB}$$

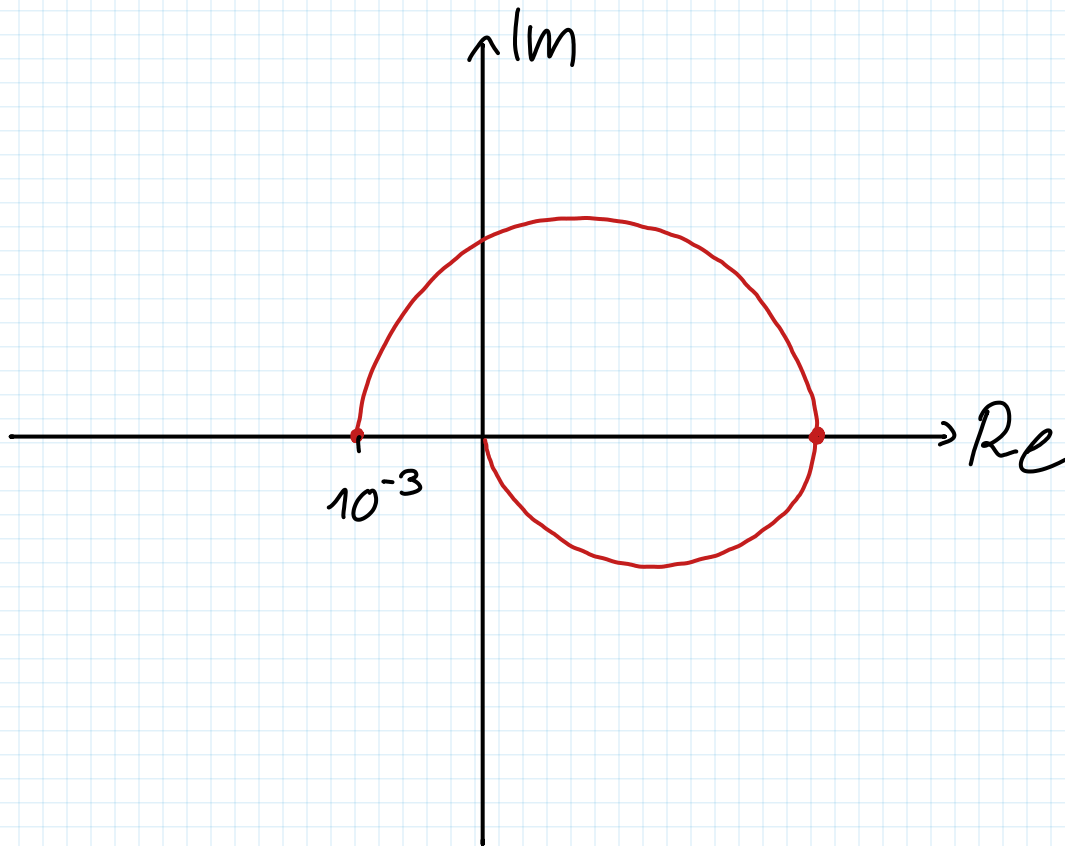
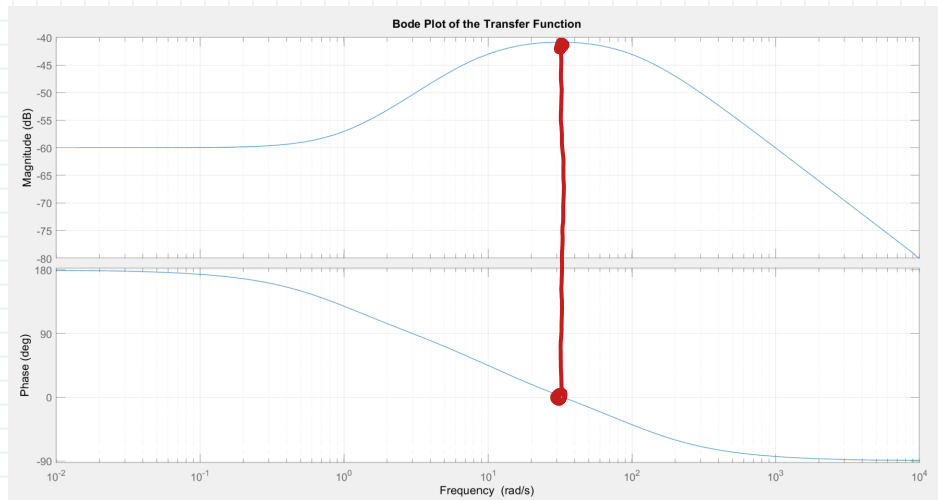
$$z_1 = 1$$

$$p_1 = -10, p_2 = -100$$

Phase \ Magnitude	-20 dB/dec	+20 dB/dec
	-90°	non-minimum phase zero
	+90°	minimum phase zero



• Now we extract the polar plot



These two plots show the same thing!
 $\Rightarrow$ Will be very important for stability analysis.

Q1) A system with a double integrator will have the following slope of the magnitude plot at $\omega(0)$:

☐ 20dB/dec

☐ -20dB/dec

☐ 40dB/dec

☒ -40dB/dec

Q2) Assume you have the Bode Plot of $G(s)$. What changes if you plot $G^{-1}(s)$?

Same plot but reflected around horizontal axis.

Q3)

Problem: Consider the transfer functions,

$$G_1(s) = \frac{s+0.1}{s^2}, \quad G_2(s) = \frac{s+1}{(s-0.1)(s+10)}, \quad G_3(s) = \frac{1}{s^2+0.02s+1}, \quad G_4(s) = \frac{s+1}{(s+0.1)(s+10)}.$$

In Figure 13 the Bode plots of the aforementioned transfer functions are shown in a randomized order.

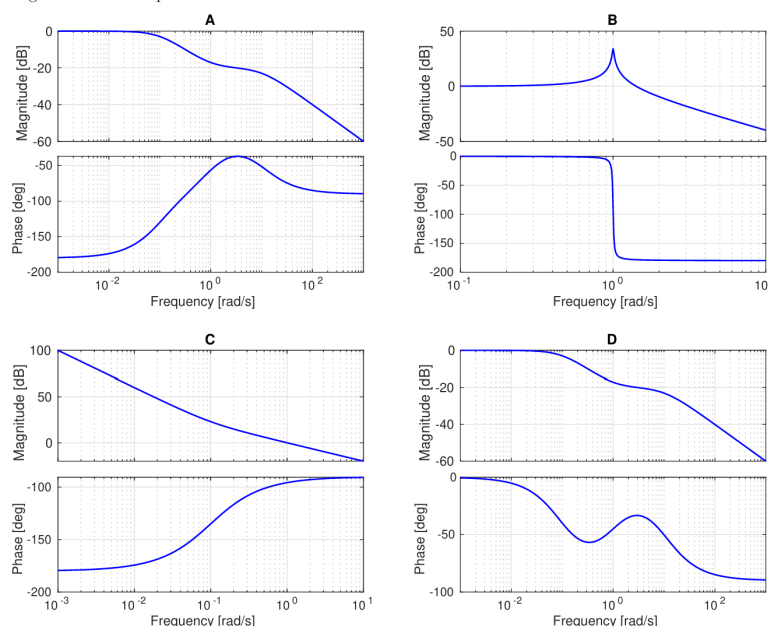


Figure 13: Bode plots of G_1 , G_2 , G_3 and G_4 in random order.

Q27 (1 Points) Mark the correct answer.

Mark the correct Bode plot and transfer function pairings?

☐ A $(A, G_1), (B, G_3), (C, G_2), (D, G_4)$

☐ B $(A, G_4), (B, G_3), (C, G_1), (D, G_2)$

☒ C $(A, G_2), (B, G_3), (C, G_1), (D, G_4)$

☐ D $(A, G_3), (B, G_2), (C, G_4), (D, G_1)$